appendix

Usage of AI

1. I used AI to clarify how I can parse the dates in different formats.

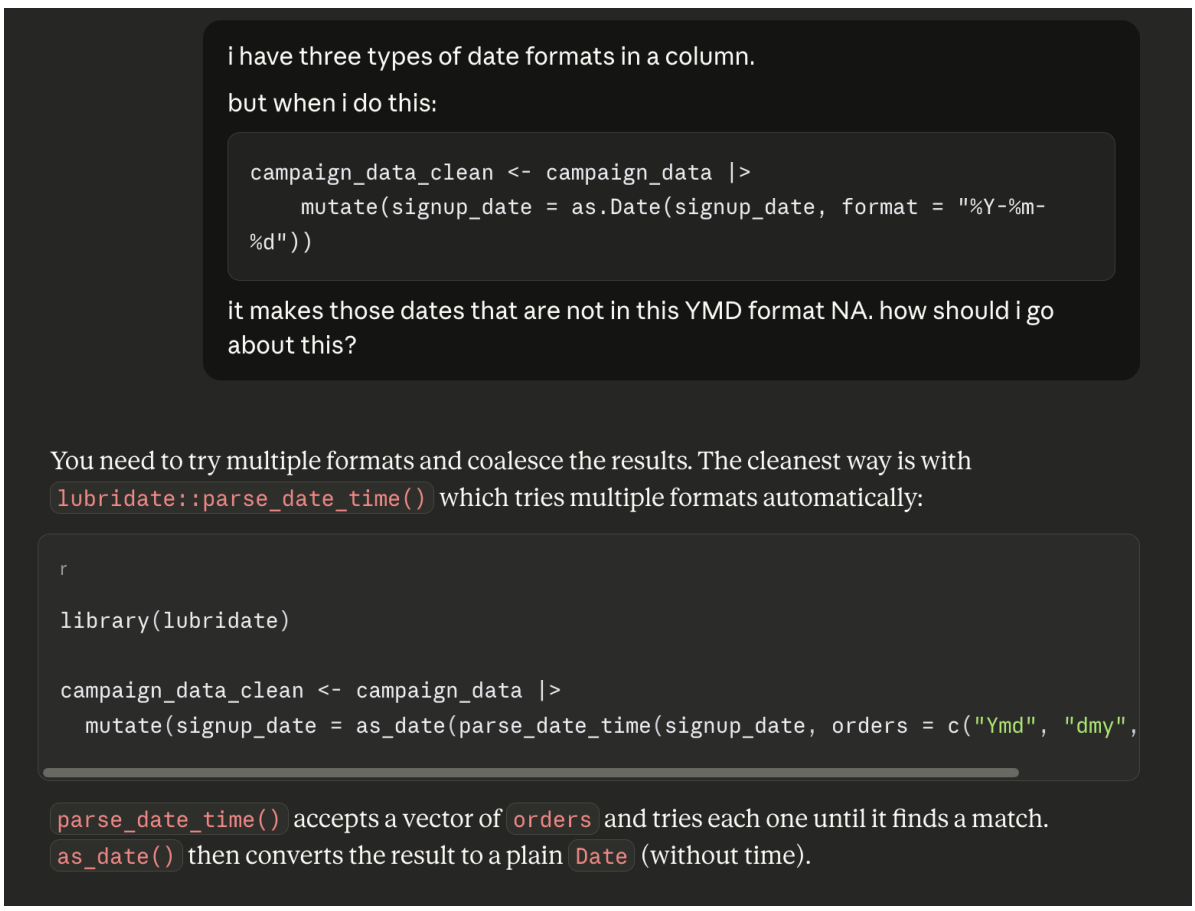

Figure 1: AI suggestion on how to use parse_date_time

2. I used AI to figure out if I would like to use boxplots, and if they are feasible.

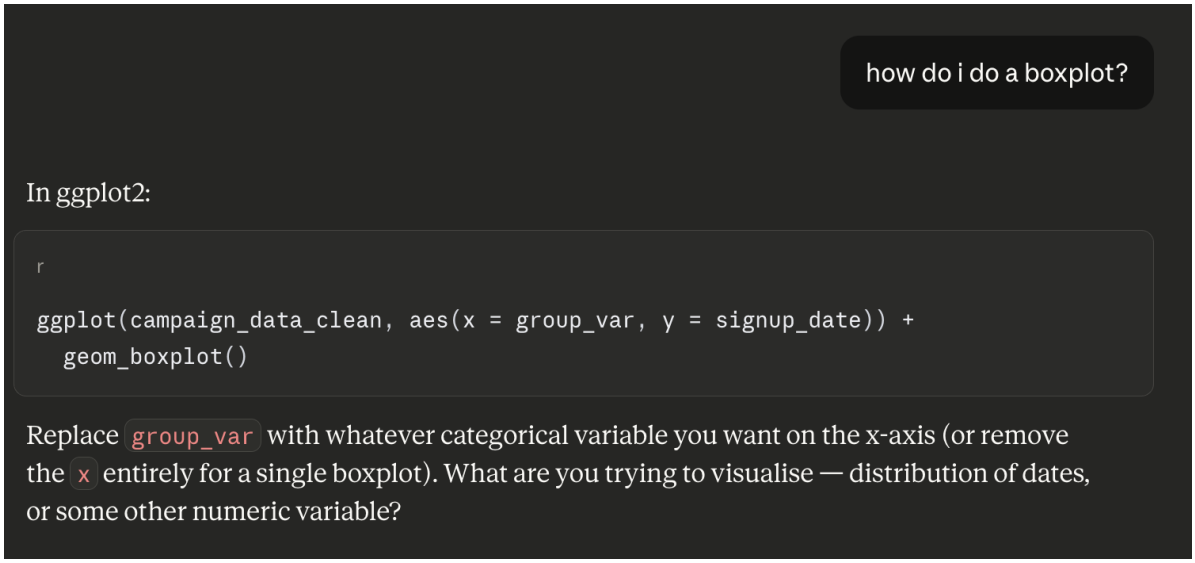

Figure 2: AI suggestion on how to use boxplot

3. Had a question on how to best sequence facet wrap with order geom functions.

```
ggplot(campaign_data_clean, aes(x = marketing_spend, y =
purchase_amount, colour = segment)) +
    facet_wrap(~segment) +
    geom_point() +
    geom_smooth(method = "lm", se = FALSE) +
    labs(title = "Figure 2. Marketing Spend vs Purchase Amount by
Segment",
         x = "Marketing Spend ($)",
```

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No difference — `facet_wrap()` is not order-sensitive relative to geoms. ggplot2 builds the plot by collecting all layers and then rendering everything together, so the position of `facet_wrap()` in the chain doesn't affect the output.

The only thing order *does* affect is geoms relative to each other — e.g. if you put `geom_smooth()` before `geom_point()`, the points will be drawn on top of the lines. Convention is to put facets/scales/themes after geoms just for readability.

Figure 3: AI suggestion on facet_wrap()